

R1 ENGINEERING CANDIDATE – NOT FOR FABRICATION
Separate replacement project: original design preserved.
USB-controlled battery power, protected ADC interfaces and isolated emitter supply.
Electrical corner review, PCB routing, assembly sourcing and physical qualification remain open.

ESP32 USB power and firmware interface

File: esp32_interface.kicad_sch

ADS1256 removable module and socket mapping

File: adc_interface.kicad_sch

Battery inputs, polarity protection and master switches

File: power_inputs.kicad_sch

Quiet held ADC supply and local digital supply

File: adc_supply.kicad_sch

USB and ESP32 supervision with battery-referenced enable

File: power_supervision.kicad_sch

Power readiness and isolation

File: power_permission.kicad_sch

SPI interface across independently switched supplies

File: spi_protection.kicad_sch

SPI defaults, decoupling and power-off leakage bounds

File: spi_defaults.kicad_sch

Detector/reference pair 1 – buffered ADC isolation

File: analog_pair_1.kicad_sch

Detector/reference pair 2 – buffered ADC isolation

File: analog_pair_2.kicad_sch

Detector/reference pair 3 – buffered ADC isolation

File: analog_pair_3.kicad_sch

Isolated adjustable emitter supply

File: emitter_regulator.kicad_sch

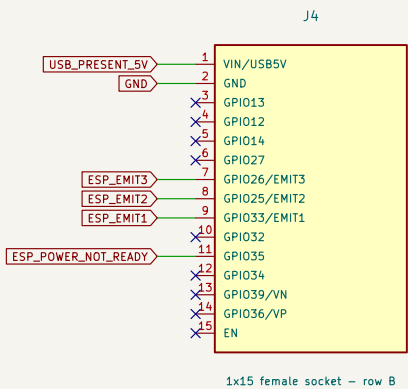
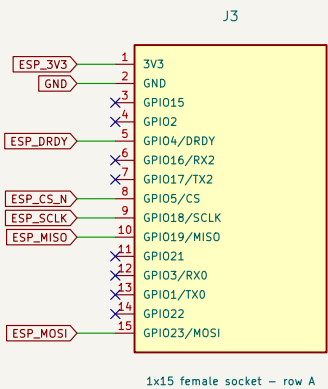
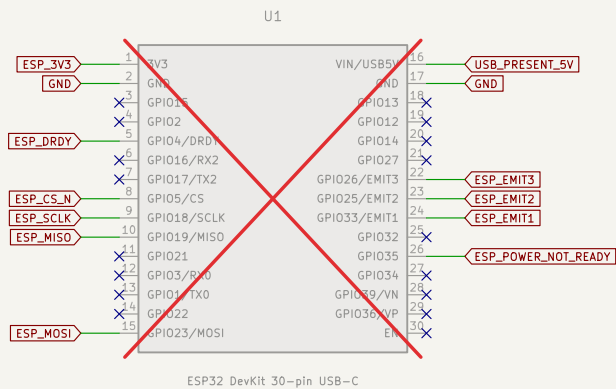
Three optically isolated emitter PWM channels

File: emitter_pwm.kicad_sch

Existing emitter wire connections

File: emitter_connections.kicad_sch

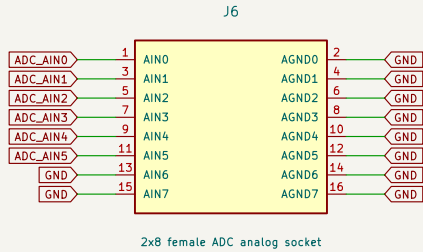
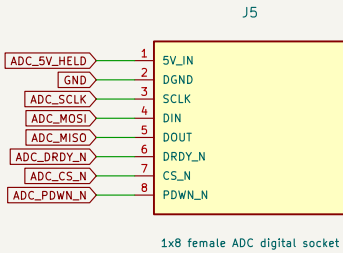
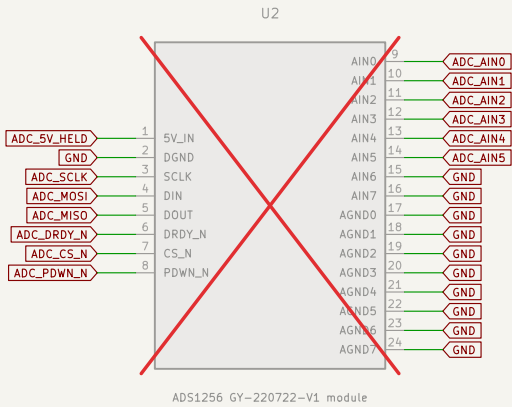
USB POWER DOMAIN
USB cable -> existing ESP32 USB-C receptacle -> module VIN pin.
No battery rail is connected to USB_PRESENT_5V or ESP_3V3.



Firmware mapping retained: SCLK18, MOSI23, MISO19, CS5, DRDY4.
Emitter outputs: GPIO33, GPIO25, GPIO26. New GPIO35 input: LOW=power ready.
SPI crosses to the switched ADC supply through powered-off-protected buffers.
Use the separate R1 firmware to wait/retry initialization and reject supply-loss captures.

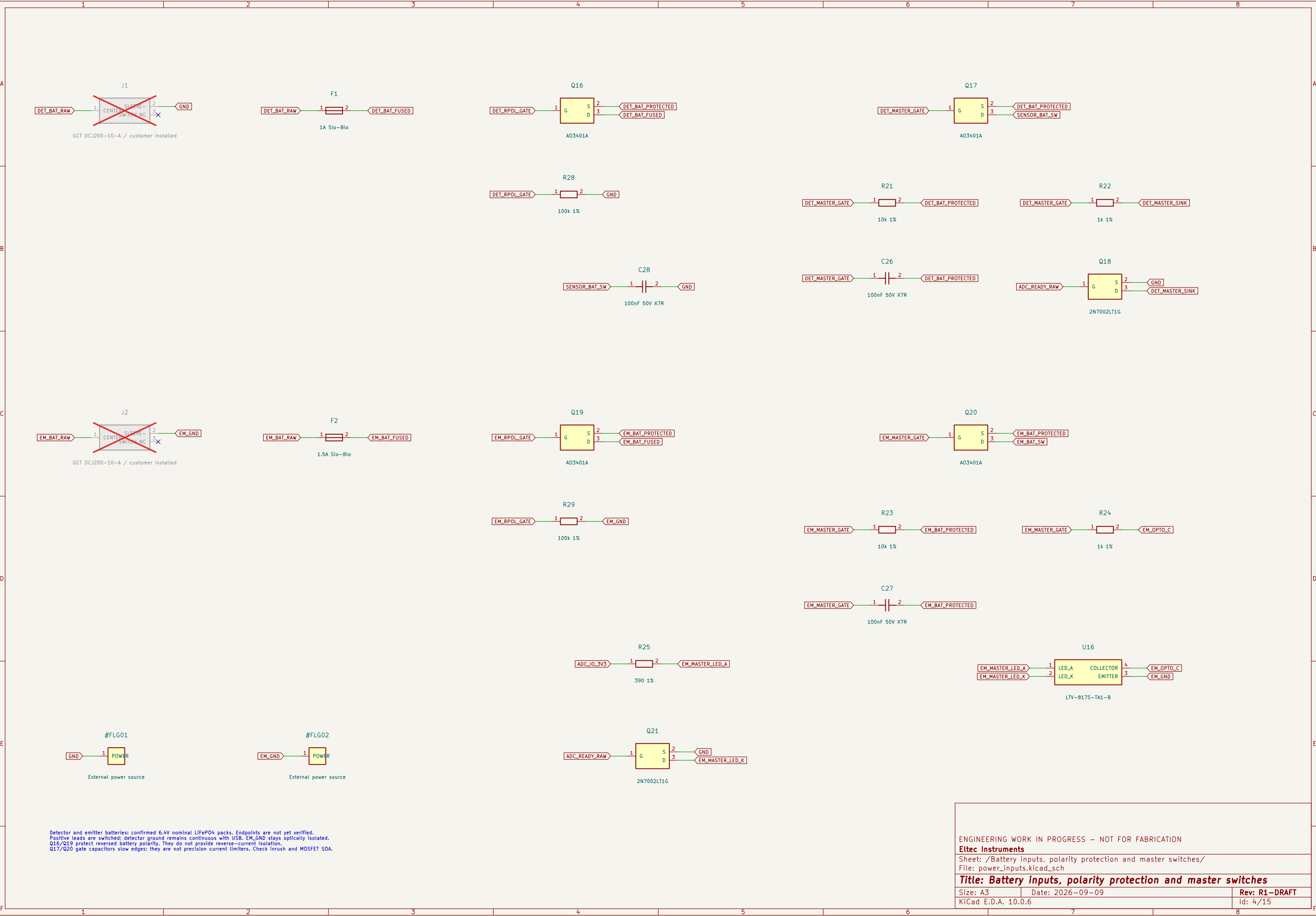
Socket row separation: 25.40 mm. Pitch: 2.54 mm.
Original duplicate pad28 is not reproduced.
Both module ground pins are explicitly connected to GND.
Only the plug-in module is omitted from PCBA; both sockets are populated.

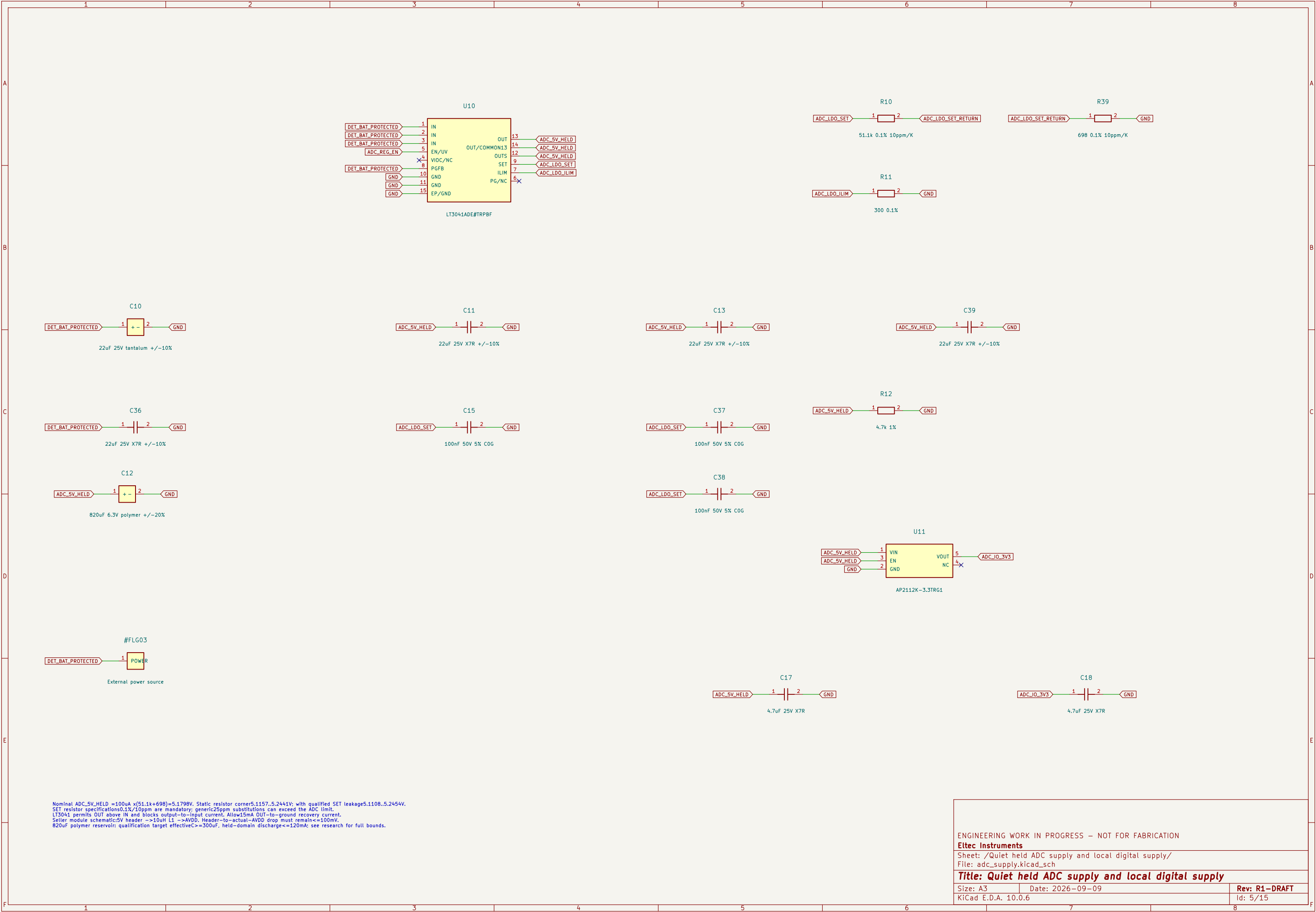
ADC BATTERY DOMAIN – seller schematic, ASIN B0DBSWRQZ5
Header5V ->10uH -> AVDD with22uF/1uF/100nF bypass; AVDD -> onboard3.3V regulator.
Each analog input has100R series and100nF at the ADC.
Verify installed module revision, filterDCR and actualAVDD before release.

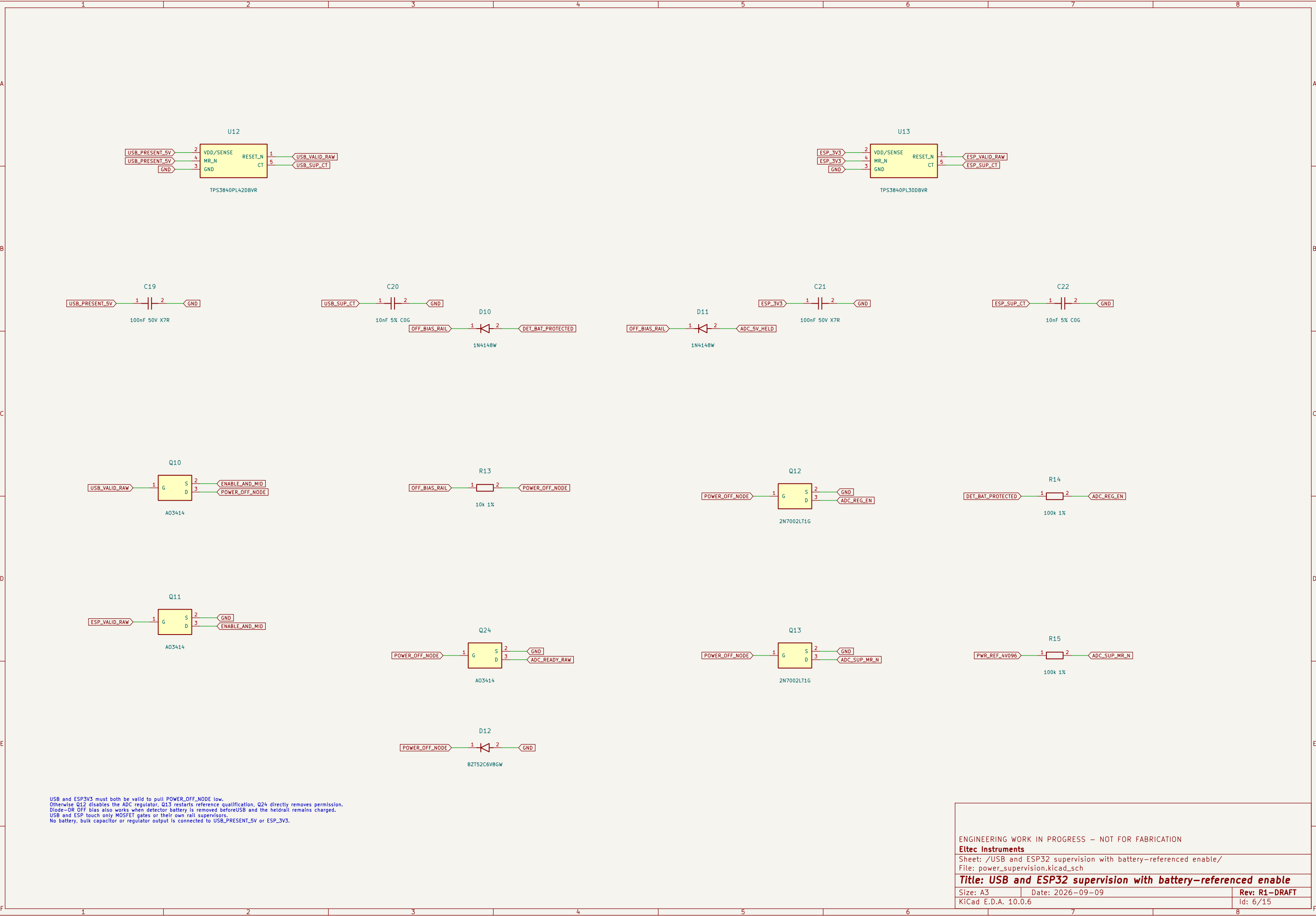


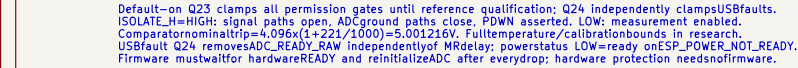
AIN0/1: test/reference pair1; AIN2/3: pair2; AIN4/5: pair3.
Unused AIN6 and AIN7 are tied to GND.
The legacy AIN7 battery divider is removed; firmware does not use it.
All eight analog grounds and digital ground are joined explicitly on the carrier.

J6 physical numbering is interleaved by channel.
This is a socket footprint, not the module body.
The build validates each socket pad against the source module coordinates.
Neither module nor barrel jacks will be included in JLC population.

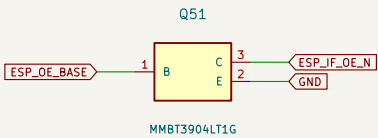
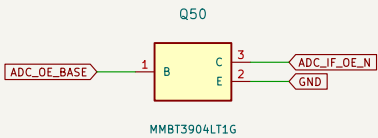
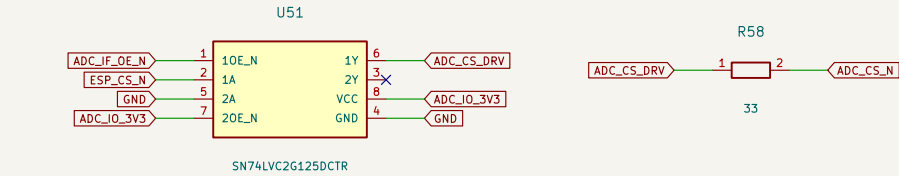
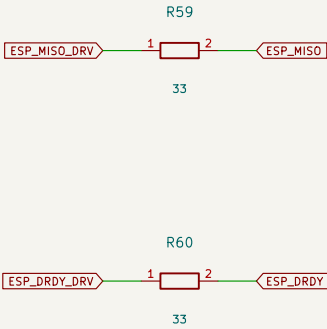
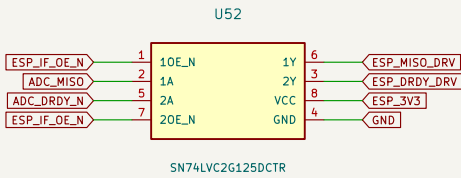
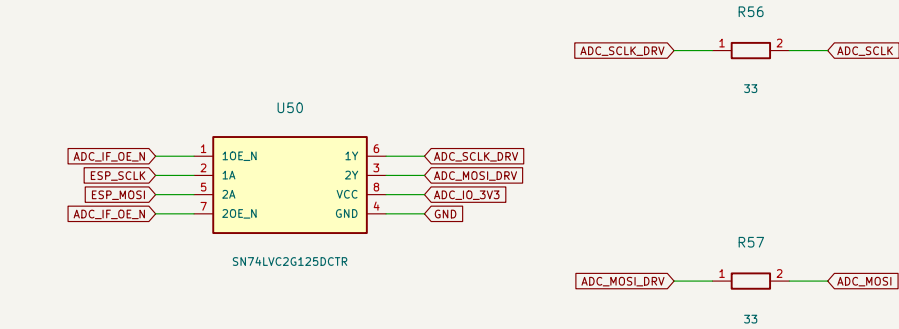




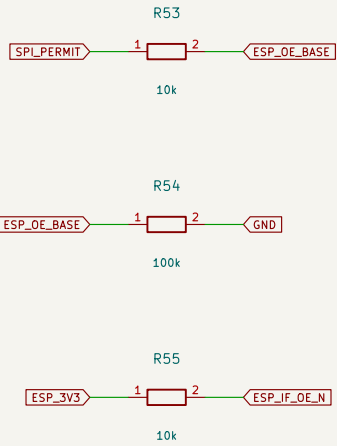
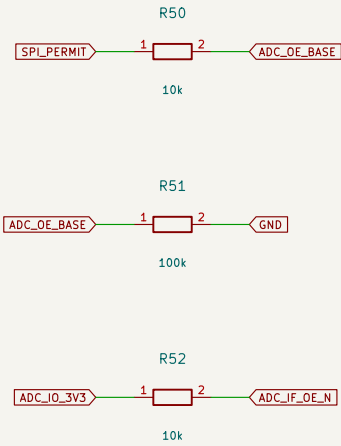




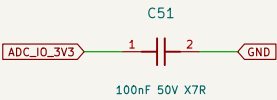
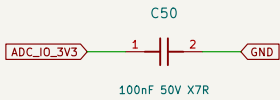
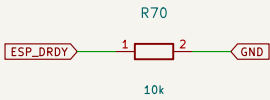
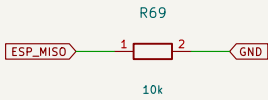
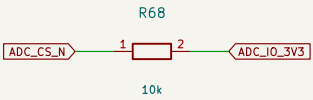
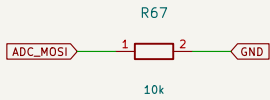
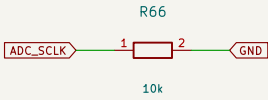
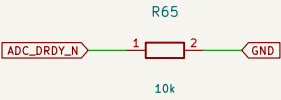
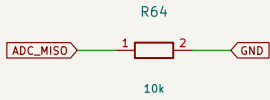
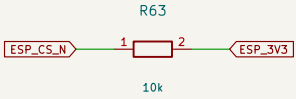
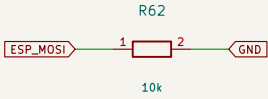
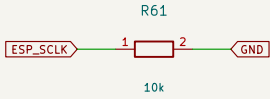
FORWARD: ADC-powered outputs. RETURN: ESP32-powered outputs.
SPL_PERMIT must be LOW until both supplies are valid; power stage provides this.
Each OE defaults HIGH through a pull-up to its own supply.



U51 spare channel: input tied LOW; OE tied HIGH; output NC.
DCT pinout: 10E=1, 1A=2, 1Y=6, 20E=7, 2A=5, 2Y=3, GND=4, VCC=8.
Do not substitute the quad SN74LVC125A: its cited data sheet does not specify loff.



RECEIVER DEFAULTS
10k pull resistors are connected to the receiver
supply or GND; CS defaults HIGH. GPIO5 boot strap remains pulled HIGH.



Ioff <=10uA with VCC=0; 10k +1% bounds pull-down rise to0.101V per output.
At VCC>=3.0V and <=0.36mA load, check ADS VIH=0.8*DVDD using the actual module DVDD.
33 ohm output damping is a starting layout value; verify SPI waveform/timing on hardware.
Each100nF capacitor belongs next to its respective U50/U51/U52 supply pins.

ENGINEERING WORK IN PROGRESS – NOT FOR FABRICATION

Eltec Instruments

Sheet: /SPI defaults, decoupling and power-off leakage bounds/
File: spi_defaults.kicad_sch

Title: SPI defaults, decoupling and power-off leakage bounds

Size: A3

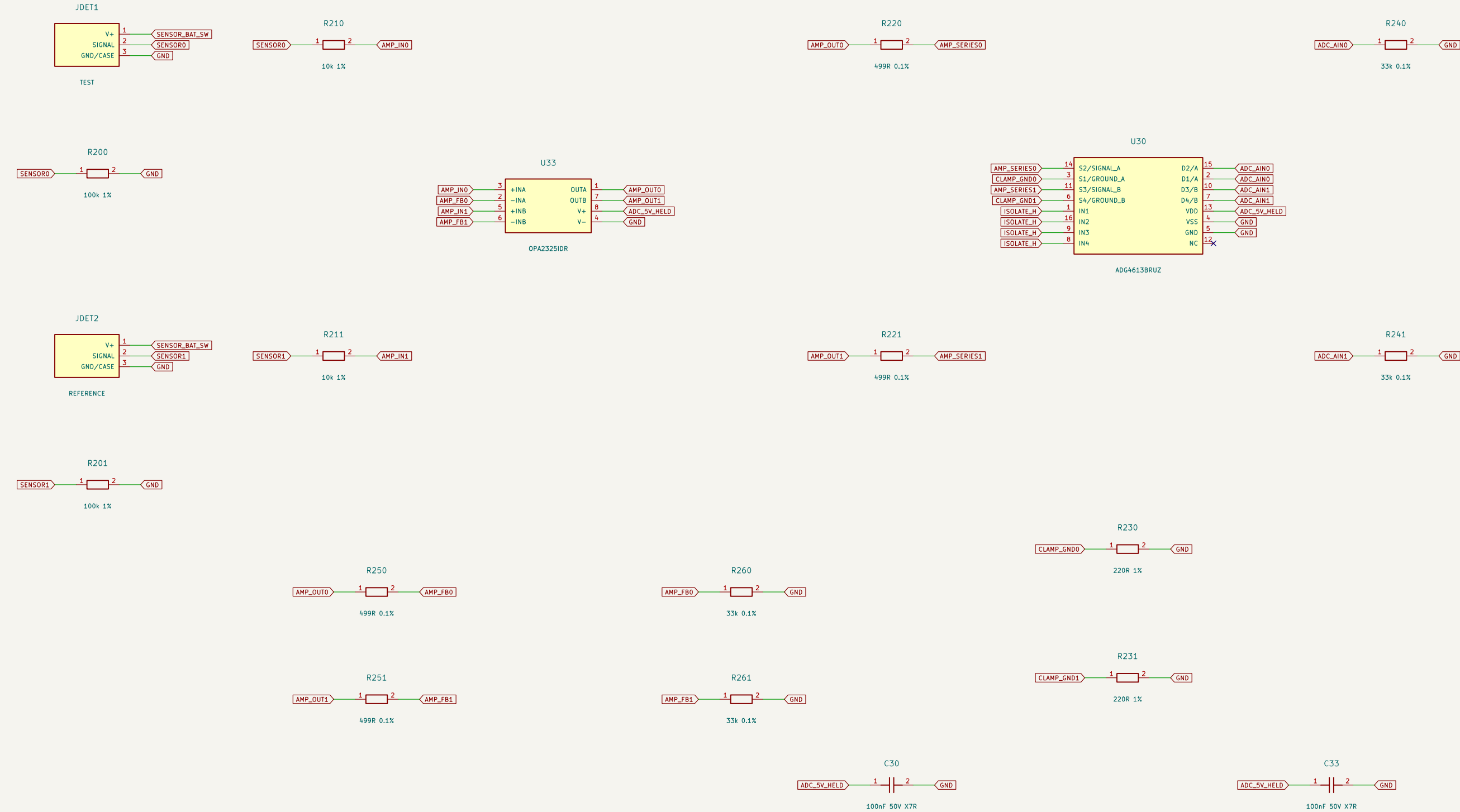
Date: 2026-09-09

Rev: R1-DRAFT

KiCad E.D.A. 10.0.6

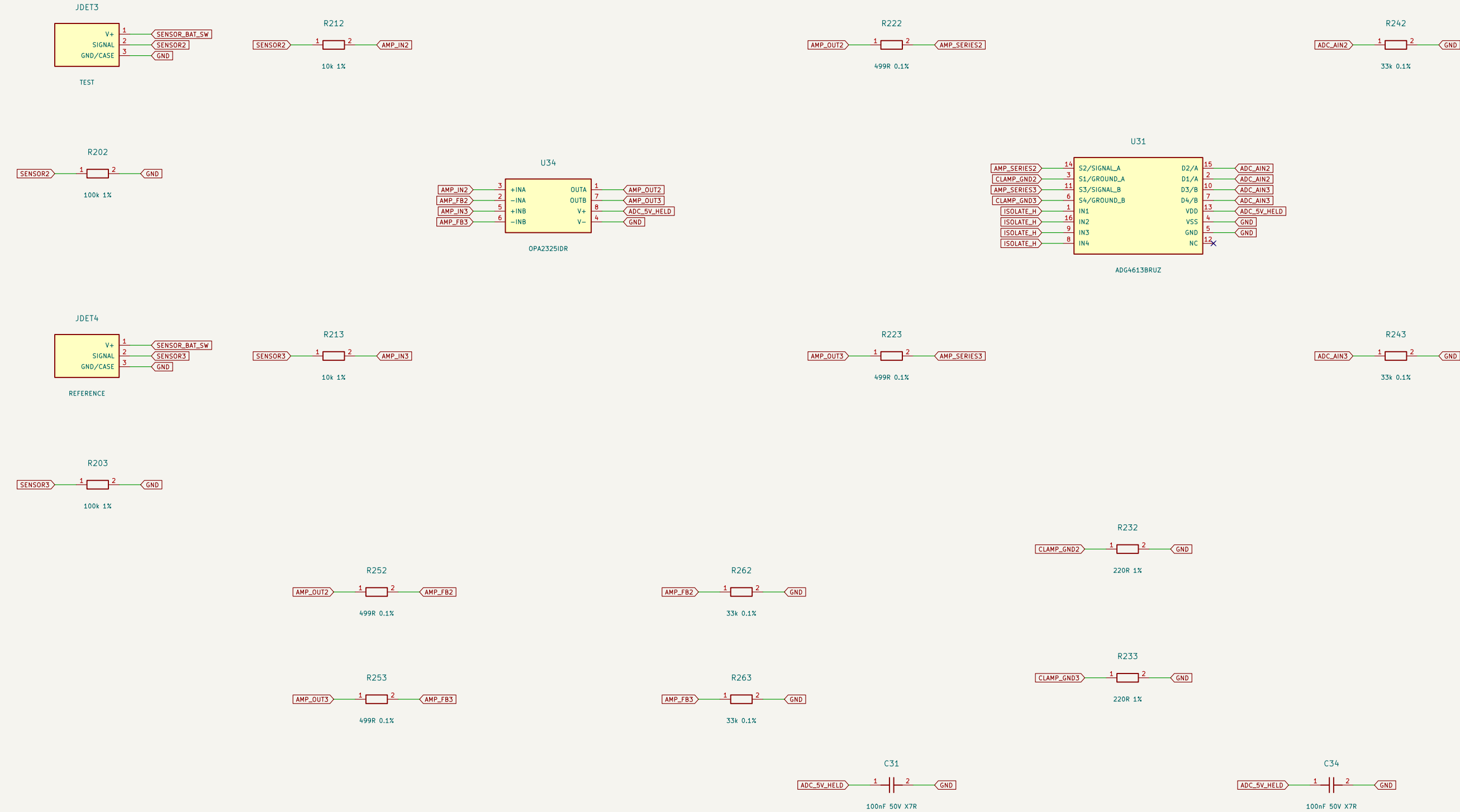
Id: 9/15

CONTROL: ISOLATE_H=1 disconnects buffers and grounds ADC inputs;0selects buffered signals.
The100k detector loads are retained. Switch leakage acts on the low-impedance buffer output.
ADC bleeders remain connected when the switches lose supply; gain compensates their nominal load.



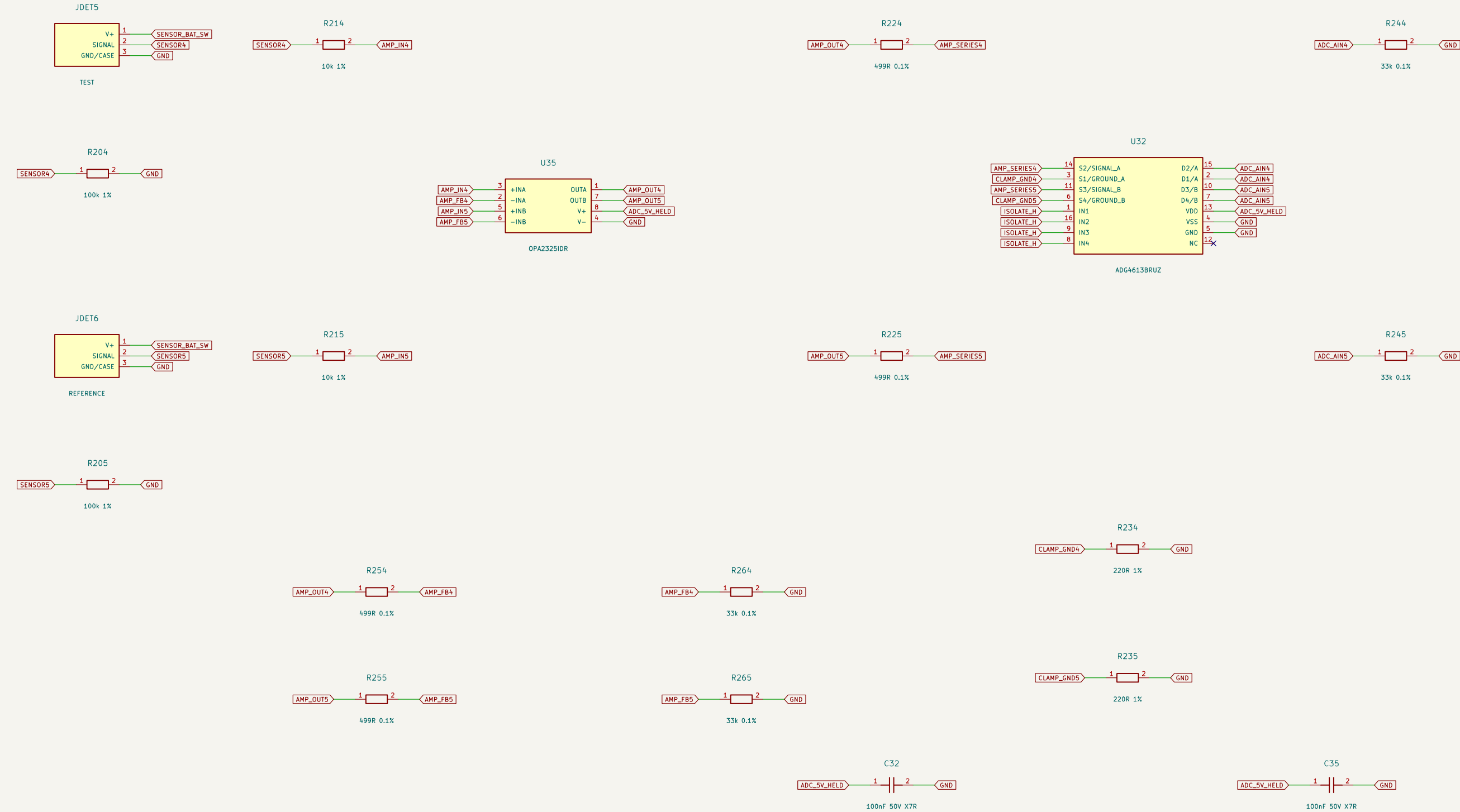
CHANNEL MAP: TEST -> AIN0; REFERENCE -> AIN1.
Held supply must outlast switch operation and module100R/100nF discharge.
The transfer includes module input impedance; this is not exact unity gain.
High faulty-detector outputs may clip; valid405 offsets0.8..3.0V remain within range.

CONTROL: ISOLATE_H=1 disconnects buffers and grounds ADC inputs;0selects buffered signals.
The100k detector loads are retained. Switch leakage acts on the low-impedance buffer output.
ADC bleeders remain connected when the switches lose supply; gain compensates their nominal load.



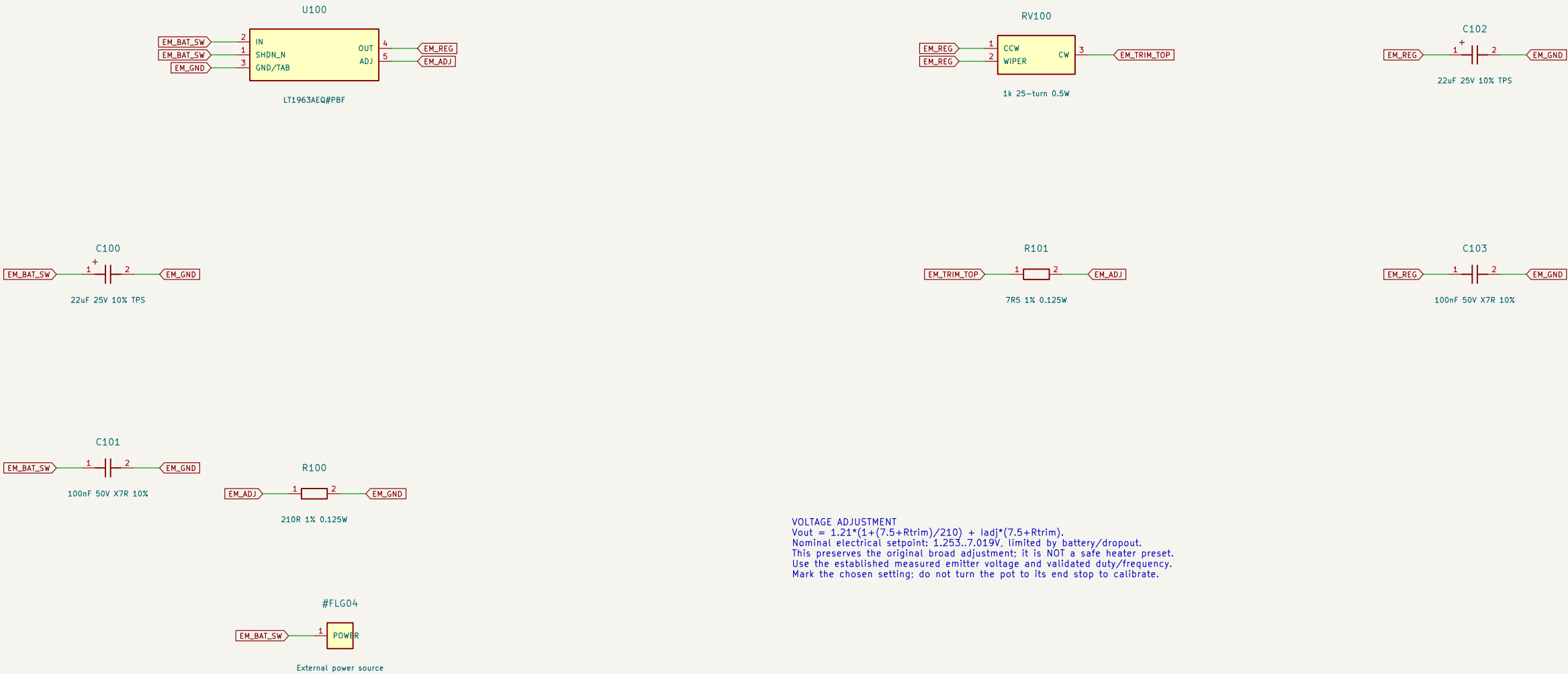
CHANNEL MAP: TEST -> AIN2; REFERENCE -> AIN3.
Held supply must outlast switch operation and module100R/100nF discharge.
The transfer includes module input impedance; this is not exact unity gain.
High faulty-detector outputs may clip; valid405 offsets0.8..3.0V remain within range.

CONTROL: ISOLATE_H=1 disconnects buffers and grounds ADC inputs;0selects buffered signals.
The100k detector loads are retained. Switch leakage acts on the low-impedance buffer output.
ADC bleeders remain connected when the switches lose supply; gain compensates their nominal load.



CHANNEL MAP: TEST -> AIN4; REFERENCE -> AIN5.
Held supply must outlast switch operation and module100R/100nF discharge.
The transfer includes module input impedance; this is not exact unity gain.
High faulty-detector outputs may clip; valid405 offsets0.8..3.0V remain within range.

SUPPLY INTERFACE
EM_BAT_SW comes from the isolated positive master switch.
SHDN_N follows EM_BAT_SW; firmware does not bypass the master.
EM_GND remains separate from GND on every copper layer.
Input bypass connects directly IN-to-EM_GND; the original D1/C5 error is removed.



VOLTAGE ADJUSTMENT
 $V_{out} = 1.21 \cdot (1 + (7.5 + R_{trim}) / 210) + I_{adj} \cdot (7.5 + R_{trim})$.
Nominal electrical setpoint: 1.253..7.019V, limited by battery/dropout.
This preserves the original broad adjustment; it is NOT a safe heater preset.
Use the established measured emitter voltage and validated duty/frequency.
Mark the chosen setting; do not turn the pot to its end stop to calibrate.

THERMAL DESIGN
Three channels: 200mA each electrical design envelope.
600mA at every trim voltage is NOT thermally qualified.
 $P = (V_{in} - V_{out}) \cdot I_{out} + V_{in} \cdot I_{gnd}$. See research/emitter_stage.md.
TO-263 tab/pin3: broad EM_GND copper and thermal vias.
No solid ground plane may bridge the optical isolation boundary.

ENGINEERING WORK IN PROGRESS – NOT FOR FABRICATION

Eltec Instruments

Sheet: /Isolated adjustable emitter supply/

File: emitter_regulator.kicad_sch

Title: Isolated adjustable emitter supply

Size: A3

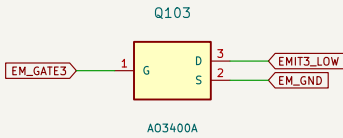
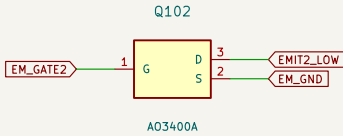
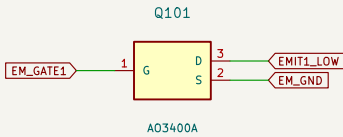
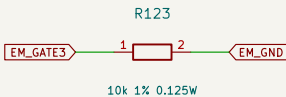
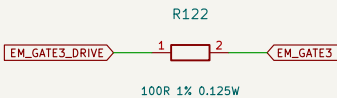
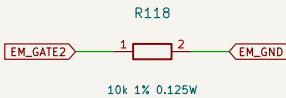
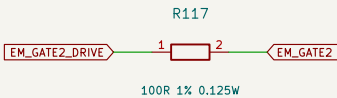
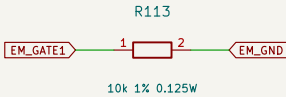
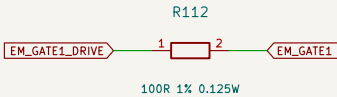
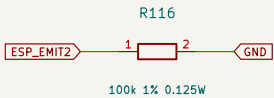
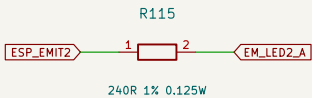
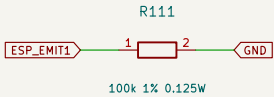
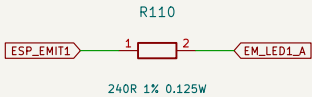
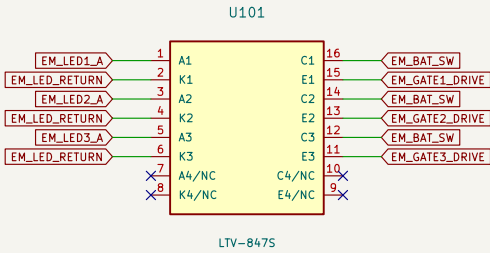
Date: 2026-09-09

Rev: R1 – DRAFT

KiCad E.D.A. 10.0.6

Id: 13/15

GPIO33 -> ESP_EMIT1 -> emitter1. GPIO25 -> emitter2. GPIO26 -> emitter3.
All outputs default off when control pins float. An asserted master permit alone does not pulse emitters.
Connect each heater positive to EM_REG; connect its return ONLY to its EMITn_LOW net.
Do not connect the switched heater return directly to EM_GND, which would bypass PWM.



GND = ESP32/USB side. EM_GND = isolated emitter battery side.
EM_LED_RETURN is sunk to GND only while the power stage permits operation.
MASTER_PERMIT therefore blocks all optical drive independently of the three GPIO commands.
Verify startup/shutdown, PWM timing, low setting gate enhancement, current and temperature.

ENGINEERING WORK IN PROGRESS – NOT FOR FABRICATION

Eltec Instruments

Sheet: /Three optically isolated emitter PWM channels/
File: emitter_pwm.kicad_sch

Title: Three optically isolated emitter PWM channels

Size: A3 Date: 2026-09-09

KiCad E.D.A. 10.0.6

Rev: R1-DRAFT

Id: 14/15

